The escape dictionary beyond one dimension: channel diffusions, switched escape processes, and the exact obstruction

Companion note to Theorem 6.1 and Secs. 8–9 of the paper
(builds on the 1D Lemma of `p1_thm_dictionary_feller_leg_proof.tex`)

Status conventions. Every statement below is tagged *[proved]*, *[proved, sketch]* (rigorous reduction to cited standard machinery, details routine but not written), or *[conjecture]*. "The 1D Lemma" means the Lemma of the companion note: for the escape diffusion $\mathrm{d}X = v(X) \circ (\beta\,\mathrm{d}t + \varepsilon\,\mathrm{d}W)$, $v = \sqrt{2(E-V)}$, on $[x_0, \infty)$ (reflected at x_0, minimal at $+\infty$), the boundary $+\infty$ is *regular* iff the classical time of flight $T = \int^{\infty} \mathrm{d}x/v$ is finite (minimal semigroup sub-stochastic and explosive, mean lifetime T^2 for $\beta = 0$, $T + O(\varepsilon^2)$ for $\beta = 1$) and *natural* iff $T = \infty$ (conservative and the unique Feller realization); the driftless speed measure is the WKB deficiency density $2[2(E-V)]^{-1/2}\mathrm{d}x$; only continuity of V and eventual $V < E$ are used.

1 What replaces the single end and the single integral

In one dimension the escaping end is a point and one integral $T(E)$ decides all three legs. In the paper's multidimensional setting the transverse dynamics is resolved into channels: for the separable model of the channel-additivity theorem (paper Thm. 8.2, `thm:channels`),

$$H \;=\; h_1 \otimes \mathbf{1} + \mathbf{1} \otimes h_\perp \quad \text{on } L^2(\mathbb{R} \times \mathbb{R}^m), \qquad h_1 = \tfrac{1}{2}p^2 + V_1, \quad h_\perp \varphi_n = E_n \varphi_n,$$

the classical escape mechanism conserves the transverse energy, and the quantum deficiency vectors are $u_n^{\pm} \otimes \varphi_n$. The correct multidimensional replacements, established below, are:

- the single escaping end $+\infty$ becomes the *countable set of channel ends* $\{\infty_n\}_{n\in\mathbb{N}}$ (one per transverse eigenstate); for continuous transverse spectrum, a measure space of fiber ends (Sec. 6);
- the time of flight becomes the *per-channel* family

  $$T_n \;=\; T(E-E_n) \;=\; \int_{x_n}^{\infty} \frac{\mathrm{d}x}{\sqrt{2\,(E - E_n - V_1(x))}}, \qquad x_n \text{ any base point with } \sup_{x \ge x_n} V_1 < E - E_n;$$

  this is the precise form taken by the anticipated "direction-dependent $T(E, \omega)$": the direction label ω is quantized into the channel label n (for central potentials, literally the angular decomposition, Corollary 6);
- and—genuinely new in $d > 1$—there is a *second* kind of end, the *channel tail* $n \to \infty$ at finite or co-moving x, which has no 1D counterpart and which is exactly where the clean dictionary stops being a theorem (Secs. 5–6).

Throughout, V_1 is continuous with $V_1(x) \to -\infty$; where the quantum leg is invoked we assume in addition the paper's Theorem 6.1 regularity. A first elementary but structurally important fact:

Lemma 1 (channel class equality; proved)**.** *If $V_1(x) \to -\infty$ then for all n, m: $T_n < \infty \iff T_m < \infty$. Hence the* qualitative *classification is channel-uniform; only the* quantitative *data $(T_n)_n$ is channel-dependent.*

Proof. $\dfrac{E - E_n - V_1(x)}{E - E_m - V_1(x)} \to 1$ as $x \to \infty$, so the integrands are comparable near $+\infty$; the finite parts are finite by continuity and the choice of base points. □

This is the stochastic shadow of the λ-independence of the limit-point/limit-circle dichotomy used in the paper's channel-additivity proof: there, every channel is limit circle because one channel is; here, every channel end is regular because one is.

2 The separable case: full per-channel generalization

The channel escape process. Define the state space as the disjoint union of half-lines

$$\mathcal{E} \;=\; \bigsqcup_{n\in\mathbb{N}} I_n, \qquad I_n = [x_n, \infty),$$

and on each I_n the channel escape diffusion $X^{(n)}$ of the 1D Lemma with speed $v_n(x) = \sqrt{2(E - E_n - V_1(x))}$, clock $\tau_n(x) = \int_{x_n}^{x} \mathrm{d}y/v_n(y)$, reflection at x_n, minimal (absorbing) convention at ∞_n; generator $\mathcal{L} = \bigoplus_n L_n$, $L_n = \frac{\varepsilon^2}{2}(v_n\partial_x)^2 + \beta\, v_n\partial_x$, $\beta \in \{0, 1\}$. This is the stochastic realization of the separable mechanism: the transverse motion is bounded oscillation whose only dynamical residue is its conserved energy, i.e. the frozen channel label.

Theorem 2 (channel dictionary, separable case; proved except (iv) countable part)**.** *Let V_1 be continuous, $V_1 \to -\infty$, and let (E_n, φ_n) be as above. Then:*

(i) ***(per-channel classification)*** *Each channel end ∞_n is* regular *iff $T_n < \infty$ and* natural *iff $T_n = \infty$ (never exit, never entrance); by Lemma 1 all channel ends are in the same class. For $\beta = 0$ the Feller test integrals in channel n are $\Sigma_n = N_n = T_n^2$.*

(ii) ***(explosion)*** *If $T_n < \infty$ (equivalently, for all n): started in channel n, the process explodes a.s. with mean lifetime T_n^2 ($\beta = 0$) resp. $T_n - \frac{\varepsilon^2}{2}(1 - e^{-2T_n/\varepsilon^2}) \in (T_n - \frac{\varepsilon^2}{2}, T_n]$ ($\beta = 1$), and $\zeta \to T_n$ a.s. as $\varepsilon \downarrow 0$:* each channel loses its mass at its own classical escape time. *The minimal semigroup $\bigoplus_n P_t^{\min,n}$ is strictly sub-stochastic on every channel. If $T_n = \infty$ for one (hence every) n, the direct sum is conservative and is the unique Feller realization of $\mathcal{L}$ with the given reflecting data.*

(iii) ***(deficiency = speed-measure data, channel by channel)*** *The driftless speed measure of channel n is the WKB density of the paper's deficiency vector $u_n \otimes \varphi_n$:*

$$m_n(\mathrm{d}x) \;=\; 2\big[2(E - E_n - V_1(x))\big]^{-1/2}\mathrm{d}x \;\approx\; 2\,|u_n^{\mathrm{WKB}}(x)|^2\,\mathrm{d}x, \qquad m_n\big((x,\infty)\big) \approx 2\int_x^\infty |u_n|^2,$$

and the scale mass of channel n near ∞_n is T_n. Thus deficiency indices (∞, ∞) correspond exactly to countably many boundary ends, each carrying finite scale and speed mass*—one Feller datum per deficiency channel.*

(iv) ***(lateral conditions = the S-matrix at infinity)*** *Suppose $T_n < \infty$ for all n, and truncate to M channels. The sub-Markov Feller extensions of the minimal process are exactly the Feller–Wentzell lateral conditions at the M regular ends: at each ∞_n a convex datum (absorption, killing, reflection, stickiness, and a jump-in measure ν_n on $\mathcal{E}$), the jump-in measures being allowed to charge* other *channels. The conservative extensions are those with no absorption/killing; a packet escaping in channel n then re-enters distributed according to ν_n—a channel-*mixing *return law, the stochastic mirror of the paper's unitary S-matrix at infinity, and different lateral data reproduce the regulator-dependence of the returned mixture seen in the paper's four-channel figure.* [Finite M: proved via Feller's lateral conditions and the Freidlin–Wentzell gluing theory; countable M: proved-sketch via Itô excursion theory, with a summability condition on the excursion entrance laws; the *completeness* claim (that all Feller extensions arise this way) is standard for finite M and labeled conjectural for countable M.]

Proof. (i)–(iii) are channelwise reductions. The direct sum has no interaction between channels, so every path quantity is computed in the channel where the process starts; the 1D Lemma applied with potential $V_1 + E_n$ (equivalently spectral shift $E \mapsto E - E_n$) gives (i) and (ii) verbatim, and Lemma 1 gives the class equality. Uniqueness in (ii): at a natural end no lateral condition exists (Feller), channel by channel, and a Feller extension of a direct sum that respects the generator must respect the direct-sum structure away from the boundary; since no boundary is accessible, the minimal realization is the only one. For (iii), the deficiency equation in channel n is $(h_1 - (\pm\mathrm{i} - E_n))u = 0$; to leading WKB order at real reference energy the modulus is $|u_n|^2 \approx [2(E - E_n - V_1)]^{-1/2}$ (the shift of the spectral parameter off the real axis affects the phase, not the leading modulus—the same convention already used in the paper's one-line identity), which is $\frac{1}{2}m_n$; the scale mass is $\int^\infty \mathrm{d}\tau_n = T_n$ as in 1D.

(iv), finite M. Each ∞_n is regular by (i). Realize $\mathcal{E}_M = \bigsqcup_{n\le M} I_n$, compactify each half-line by its end, and consider the star obtained by the disjoint boundary set $\{\infty_1, \dots, \infty_M\}$. Feller's classification of lateral conditions at a regular boundary of a generalized second-order operator [Feller, Illinois J. Math. 1 (1957) 459–504] provides, for a single interval, the general boundary datum including a jump-in measure on the interior; the multi-interval (graph) case, with boundary points allowed to communicate, is the Freidlin–Wentzell gluing theory [Freidlin and Wentzell, Ann. Probab. 21 (1993) 2215–2245], which parametrizes all Feller processes agreeing with the minimal one in the interior by Wentzell data at the vertices; jump-in to interior points is supplied by the Itô excursion construction of instantaneous-return processes [Itô, Proc. Sixth Berkeley Symp. (1972) 225–239; Blumenthal, *Excursions of Markov Processes*, Birkhäuser 1992]: at a regular end with finite scale and speed mass the minimal process killed at the end has an exit system, and any sub-probability kernel $(\nu_n)_{n\le M}$ together with nonnegative sticky/killing weights determines a unique right process extending it. Countable M: the same excursion construction applies end by end; what must be checked is that the concatenation over countably many ends does not accumulate infinitely many boundary visits in finite time, for which a uniform lower bound on mean return times (automatic when $\inf_n T_n > 0$, and in general a summability condition on the entrance laws) suffices. We have not written this in full generality; the parametrization is asserted as complete only for finite M. □

Remark 3 (the mirror is structural, not a bijection; honest). The quantum completions form the unitary group $U(\mathcal{K}_+, \mathcal{K}_-)$, $\dim \mathcal{K}_\pm = \infty$; the conservative stochastic completions form the convex set of Markov gluing data (ν_n). As already in 1D ($U(1)$ versus Feller's convex family), these are *parallel infinite-dimensional families attached to the same boundary objects*, in canonical

correspondence at the level of "which channels communicate with which through infinity", not isomorphic as spaces. In particular the paper's statement that conservative completions "scramble the transverse state with data injected at infinity" has an exact stochastic counterpart: the returned channel mixture is the jump-in kernel, and it is extension data, not data of $\mathcal{L}$.

Remark 4 (conservative / transient / explosive trichotomy). The paper's Sec. 5 hierarchy is reproduced by the clock picture: in channel n the drifted model is Brownian motion with unit drift on $[0, T_n)$; $T_n = \infty$ with $\beta = 1$ is *conservative and transient* (the quadratic-ghost cascade regime, paper Thm. `thm:quadratic`), $T_n < \infty$ is *explosive*, and a confining channel (no escape interval) is *recurrent*. All three currencies appear channelwise.

Remark 5 (two-ended channels). If h_1 is limit circle at both ends (the paper's `thm:channels` hypothesis), each channel contributes two regular ends and the finite-M boundary set has $2M$ points—the exact mirror of the $(2M, 2M)$ indices in the paper's finite-channel stability theorem (`thm:stability`). All statements above apply verbatim with the reflecting base replaced by the second end's data.

3 Central potentials: the sphere at infinity, resolved into channels

Corollary 6 (central case; proved)**.** *Let $H = -\frac{1}{2}\Delta - c|x|^\alpha$ on $\mathbb{R}^d$, $d \geq 2$, $c > 0$, $\alpha > 2$ (the paper's Corollary 8.3). In the angular decomposition the radial channel ℓ carries the escape diffusion with $v_\ell(r)^2 = 2\big(E + cr^\alpha - L^2/(2r^2)\big)$, $L^2 = (\ell + \frac{d-2}{2})^2 - \frac{1}{4}$, on $r \geq r_\ell$ beyond the outer turning point. Then every channel end is regular, with*

$$T_\ell \asymp C_\alpha\, c^{-\frac{2}{\alpha+2}}\, L^{\frac{2-\alpha}{2+\alpha}} \xrightarrow[\ell\to\infty]{} 0, \qquad C_\alpha = \int_{\rho_0}^{\infty} \big(2(\rho^\alpha - \tfrac{1}{2}\rho^{-2})\big)^{-1/2} \mathrm{d}\rho < \infty,$$

so $T^* = \sup_\ell T_\ell < \infty$: the classification is uniform across the entire sphere of directions. All conclusions of Theorem 2 hold with n replaced by ℓ (finite multiplicities absorbed into the channel count); in particular the conservative completions of the central malicious ghost are angular-momentum-mixing return laws at $r = \infty$.

Proof. Substituting $r = \lambda\rho$ with $\lambda = (L^2/c)^{1/(\alpha+2)}$ equalizes the potential and centrifugal scales, $cr^\alpha = c\lambda^\alpha\rho^\alpha$, $L^2/(2r^2) = c\lambda^\alpha\rho^{-2}/2$, giving $T_\ell = \lambda(2c\lambda^\alpha)^{-1/2}\big(C_\alpha + o(1)\big) = (2c)^{-1/2}\lambda^{1-\alpha/2}(C_\alpha + o(1))$ and $\lambda^{1-\alpha/2} = (L^2/c)^{\frac{2-\alpha}{2(\alpha+2)}}$. (Numerical check, $\alpha = 4$, $c = 1$: T_ℓ at $L^2 = 10^2, 10^4, 10^6$ is 1.194, 0.554, 0.257, successive ratios 2.155 against the predicted $10^{1/3} = 2.154$.) The ρ-integral converges at infinity because $\alpha > 2$ and at the turning point because the turning point of $\rho^\alpha - \frac{1}{2}\rho^{-2}$ is simple. The rest is Theorem 2. □

Thus for central potentials the anticipated "per-direction $T(E, \omega)$" collapses, correctly, to the channel family T_ℓ: the boundary at infinity is not the geometric sphere S^{d-1} but its harmonic resolution, and the uniformity $T^* < \infty$ is automatic. This is the multidimensional situation in which the generalization is *complete*.

4 Bounded channel mixing: the stochastic Kato theorem

The paper's non-separable theorem (`thm:kato`) says a bounded mixing $B = \mu f(x_1)g(x_2)$ preserves the deficiency indices. The stochastic counterpart replaces the frozen channel label by a jump process with *bounded intensity*.

The switched escape process. Fix measurable rates $\lambda(x; n \to m) \geq 0$ with $\sup_{x,n} \sum_m \lambda(x; n \to m) \leq \kappa < \infty$, supported on *open* channels ($\lambda(x; n \to m) = 0$ unless $x \geq x_m$). Realize the pair (X_t, n_t) on a probability space carrying the driving Brownian motion W and an independent Poisson process of intensity κ with i.i.d. uniform marks: between candidate times the path solves the frozen channel SDE $\mathrm{d}X = v_n(X) \circ (\beta\,\mathrm{d}t + \varepsilon\,\mathrm{d}W)$ (reflected at x_n); at a candidate time a switch $n \to m$ is accepted with probability $\lambda(X; n \to m)/\kappa$ (thinning). X is continuous across switches.

Theorem 7 (stability under bounded mixing; proved)**.** *For the switched escape process:*

(a) ***(conservative side, no extra hypothesis)*** *If $T_n = \infty$ for one (hence every) channel, the switched process is conservative: $\mathbb{P}(\zeta < \infty) = 0$ for every initial condition and* every *admissible bounded-rate switching law.*

(b) ***(explosive side, uniform time of flight)*** *If*

$$\text{(U)}: \qquad T^* := \sup_n T_n < \infty,$$

then the switched process explodes a.s. for every admissible bounded-rate law, with exponential tail: for $\beta = 1$,

$$\mathbb{P}(\zeta > kh) \leq (1 - p_0)^k, \qquad h = T^* + 1, \quad p_0 = e^{-\kappa h}\,\mathbb{P}\Big(\varepsilon \sup_{s \leq h} |W_s| \leq \tfrac{1}{2}\Big) > 0,$$

so $\mathbb{E}[\zeta] \leq h/p_0 < \infty$; the same holds for $\beta = 0$ with $p_0 = e^{-\kappa h}\mathbb{P}(\sup_{s \leq h} W_s \geq T^*)$.

Hence under (U) the dictionary survives bounded mixing verbatim: $T < \infty$ class $\Leftrightarrow$ explosive, $T = \infty$ class $\Leftrightarrow$ conservative—the stochastic mirror of Kato stability of the deficiency indices.

Proof. (a) The candidate switch times form a Poisson(κ) process independent of everything else, so a.s. only finitely many candidates (hence switches) occur in $[0, t]$. If $\zeta \leq t$, the explosion instant lies in some inter-candidate interval $[\sigma_k, \sigma_{k+1})$; on that interval the path is the frozen channel-n_k diffusion started at the finite point X_{σ_k}, whose end ∞_{n_k} is natural (1D Lemma with $T_{n_k} = \infty$), hence inaccessible in finite time—a null event. A countable union over k of null events is null.

(b) Work on windows $[kh, (k+1)h)$. Let A_k be the event that no Poisson candidate falls in the window and $\varepsilon \sup_{s \leq h} |W_{kh+s} - W_{kh}| \leq \frac{1}{2}$; by independence of the Poisson clock and the Brownian increments from $\mathcal{F}_{kh}$, $\mathbb{P}(A_k \mid \mathcal{F}_{kh}) \geq p_0$. On A_k, if $\zeta > kh$, the process sits in some channel n at clock position $u = \tau_n(X_{kh}) \in [0, T_n)$ and, with no switch in the window, follows the frozen dynamics; in the clock coordinate (1D Lemma proof) the Skorokhod decomposition gives $Z_s \geq u + s - \frac{1}{2} \geq s - \frac{1}{2}$, and $s = T_n + \frac{1}{2} \leq T^* + \frac{1}{2} < h$ already forces Z to reach T_n: explosion occurs inside the window. Hence $\mathbb{P}(\zeta > (k+1)h \mid \zeta > kh) \leq 1 - p_0$ and the geometric bound follows; positivity of p_0 is elementary. For $\beta = 0$ the frozen clock process is reflected Brownian motion, and $\sup_{s \leq h} W_s \geq T^*$ forces the hit; the reflection only helps. Note the argument never needs to know *which* channels are visited—only (U) and the open-channel support of the rates (which guarantees the frozen dynamics is defined after every switch). □

Lemma 8 (when (U) is automatic; proved)**.** *Suppose, for $x \geq R$: $-V_1(x) \geq c_1 x^{2+\epsilon}$ and $-V_1'(x) \geq c_3 x^{1+\epsilon}$ with $\epsilon > 0$, $V_1 \in C^1$ strictly decreasing there, and $T(E) < \infty$. Then $\sup_{E' \leq E} T(E') < \infty$; in particular (U) holds for every choice of transverse spectrum (E_n) bounded below or not. This covers all power-law potentials $-V_1 \sim cx^{2+\epsilon}$, hence the paper's representative models.*

Proof. Write $\mu = -E'$ ($\to \infty$), turning point $x'(E')$, and $x_g = (\mu/c_1)^{1/(2+\epsilon)} \geq x'(E')$ (since $E' + c_1 x'^{2+\epsilon} \leq E' - V_1(x') = 0$). Beyond x_g, $E' - V_1 \geq c_1 x^{2+\epsilon} - \mu$ and the scaling $x = x_g u$ gives $\int_{x_g}^{\infty} \leq C\,\mu^{-\epsilon/(2(2+\epsilon))} \to 0$ exactly as in the central computation. Between x' and x_g, the derivative bound gives $E' - V_1(x) = \int_{x'}^{x}(-V_1') \geq c_3 x'^{1+\epsilon}(x - x')$, so $\int_{x'}^{x_g} \leq \sqrt{2(x_g - x')/(c_3 x'^{1+\epsilon})} \leq C'\,x_g^{-\epsilon/2} \to 0$ (using $x' \asymp x_g$). So $T(E') \to 0$ as $E' \to -\infty$, and on any compact range of E', T is finite and continuous (simple turning points by strict monotonicity, dominated convergence with the local $(x - x')^{-1/2}$ bound). Hence the sup is finite. □

5 The obstruction: unbounded channel transfer ("channel riding")

Everything so far generalizes. What follows is the exact point of failure, and it is proved by construction, not merely feared.

Lemma 9 (channel-riding reduction; proved)**.** *Let $n(\cdot) : [x_*, \infty) \to \mathbb{N}$ be measurable and locally constant off a discrete set, with channel $n(x)$ open at x for every x. The* position-locked *switched process (switch to channel $n(x)$ upon reaching x; a limit of bounded-rate laws with rates $\to \infty$, so* outside *Theorem 7's hypothesis) is the one-dimensional generalized diffusion on $[x_*, \infty)$ with effective speed*

$$v_{\text{eff}}(x) \;=\; \sqrt{2\big(E - E_{n(x)} - V_1(x)\big)},$$

defined by scale $s_{\text{eff}}(x) = \int^x \mathrm{d}y/v_{\text{eff}}$ (for $\beta = 0$; the drifted variant analogously) and speed measure $m_{\text{eff}} = 2\,\mathrm{d}x/v_{\text{eff}}$. Its end at $+\infty$ is classified by the 1D Lemma with the effective time of flight

$$T_{\text{eff}} \;=\; \int_{x_*}^{\infty} \frac{\mathrm{d}x}{\sqrt{2\,(E - E_{n(x)} - V_1(x))}}\,.$$

Proof. Scale-and-speed constructions require only a strictly increasing continuous scale and a Radon speed measure (Itô–McKean §4); v_{eff} is measurable, locally bounded and locally bounded away from 0 on the open region, so s_{eff} and m_{eff} qualify, and the resulting regular diffusion coincides with the switched process between switch points by locality of the generator. The classification integrals are computed exactly as in the 1D Lemma ($\Sigma = N = T_{\text{eff}}^2$ for $\beta = 0$). □

T_{eff} is a *path functional of the coupled mechanism*, not a function of the per-channel family $\{T_n\}$. The next two propositions make the underdetermination rigorous, in both directions.

Proposition 10 (Counterexample I: all channels explosive, coupled process conservative; proved)**.** *There exist V_1 (continuous, strictly decreasing to $-\infty$, satisfying the paper's Theorem 6.1 regularity, including $|V_1'||V_1|^{-3/2} \to 0$) and a transverse spectrum (E_n), such that*

$$\sup_n T_n \;\leq\; \tfrac{3}{2} \qquad (\textit{every frozen channel explosive, uniformly—(U) holds}),$$

yet the position-locked riding law $n(x)$ below gives $T_{\text{eff}} = \infty$: the coupled process is conservative*. Consequently the per-channel dictionary ("$T_n < \infty\ \forall n \Rightarrow$ explosive") is* false *for unbounded-rate channel transfer, even under (U); the bounded-rate hypothesis of Theorem 7 cannot be dropped.*

Proof. Take $E = 0$, $E_n = 16^n$, and the shelf potential

$$V_1(x) = -E_n - \tfrac{1}{2} \text{ on } [2n,\, 2n+1], \qquad V_1 \text{ linear on } [2n+1,\, 2n+2], \qquad n \geq 1,$$

(extended arbitrarily and boundedly to the left; tilt the shelves by an immaterial slope if strict monotonicity is wanted). Regularity: $|V_1'| \asymp 16^n$ on the n-th drop while $|V_1|^{3/2} \asymp 16^{3n/2}$, so $|V_1'||V_1|^{-3/2} \to 0$. Channel m is open exactly from its own shelf onward ($E - E_m - V_1 = E_n + \frac{1}{2} - E_m > 0$ iff $n \geq m$), so $x_m = 2m$, and

$$T_m \;\leq\; \underbrace{\frac{1}{\sqrt{2 \cdot \frac{1}{2}}}}_{\text{own shelf}} + \underbrace{1}_{\text{own drop, speed}^2 \geq 1} + \sum_{n>m} \Big(\underbrace{\frac{1}{\sqrt{2(E_n - E_m)}}}_{\text{shelf } n} + \underbrace{\frac{2}{\sqrt{E_n}}}_{\text{drop } n} \Big) \;\leq\; \tfrac{3}{2}$$

for m large (numerically $T_1 = 1.17$, $T_3 = 1.01$, $T_8 = 1.00004$; all ≤ 1.18). Now ride: $n(x) = n$ on $[2n, 2n+2)$ (always open). On shelf n the effective speed is $\sqrt{2(E_n + \frac{1}{2} - E_n)} = 1$ over unit length, so each shelf contributes exactly 1 to T_{eff}: $T_{\text{eff}} \geq \sum_n 1 = \infty$. Lemma 9 and the 1D Lemma ($T_{\text{eff}} = \infty \Rightarrow$ natural, conservative) finish. □

Proposition 11 (Counterexample II: all channels conservative, coupled process explosive—ghost-specific; proved)**.** *Let $V_1 \equiv 0$ on $[1, \infty)$ (or $V_1 = -x^2$, the marginal case), $E = 1$, and let $h_\perp$ be a* ghost *oscillator, $E_n = -(n + \frac{1}{2})$, unbounded below. Every frozen channel has constant (or at-most-linearly-growing) speed, hence $T_n = \infty$ for all n: every frozen channel is conservative. The riding law $n(x) = \lceil x^{2+\epsilon} \rceil$ gives $v_{\text{eff}}(x)^2 = 2(E + n(x) + \frac{1}{2}) \geq 2x^{2+\epsilon}$, so $T_{\text{eff}} \leq \int^\infty x^{-1-\epsilon/2} \mathrm{d}x/\sqrt{2} < \infty$: the coupled process is* explosive*. The mechanism extracts unbounded energy from the transverse ghost—the stochastic realization of the paper's "channel-tail escape" ($\lambda < 0$ PU climbing a rising potential through arbitrarily high mode-2 channels) and of its remark that transverse observables are not H_0-bounded for indefinite $h_\perp$.*

Proof. Immediate from Lemma 9 and the 1D Lemma. □

Theorem 12 (ghosts are necessary for Counterexample II; proved under (D))**.** *Suppose $h_\perp$ is bounded below, $E_n \geq E_0$, let $\bar{v}(x)^2 = 2(E - E_0 - V_1(x))$ dominate every channel speed, and assume*

$$\text{(D)}: \qquad \int^\infty \frac{\mathrm{d}x}{\bar{v}} = \infty \quad \textit{and} \quad \limsup_{x \to \infty} \bar{v}'(x) < \infty \;\; \big(\textit{i.e. } -V_1' \leq C\,\bar{v} \textit{ eventually; holds for } V_1 \geq -Kx^2\big).$$

Then every *switched escape process—arbitrary adapted switching with countably many switches, including position-locked riding and any unbounded rates—is conservative. In particular, below-quadratic (marginal or slower) escape ends with non-ghost transverse dynamics are* absolutely *stable under channel coupling: the conservative leg of the dictionary is robust, the explosive leg is not.*

Proof. Khasminskii test with the dominating clock. Let $\bar{\tau}(x) = \int_{x_*}^x \mathrm{d}y/\bar{v}$ and let φ equal $\bar{\tau}$ for $x \geq x_* + 1$, be smoothed to constant near the base so that $\varphi' \geq 0$ vanishes at every reflection point; $\varphi \uparrow \infty$ by (D). For any channel n open at x (so $v_n(x) \leq \bar{v}(x)$), on $x \geq x_* + 1$: $v_n \varphi' = v_n/\bar{v} \leq 1$, and, using $v_n v_n' = -V_1' = \bar{v}\,\bar{v}'$ (all channels share $-V_1'$),

$$v_n\,(v_n \varphi')' = \frac{v_n v_n'}{\bar{v}} - \frac{v_n^2\,\bar{v}'}{\bar{v}^2} = \bar{v}' \Big(1 - \frac{v_n^2}{\bar{v}^2}\Big) \;\in\; \big[\min(0, \bar{v}'),\; \max(0, \bar{v}')\big] \;\leq\; C.$$

Hence $L_n\varphi \le \beta + \frac{\varepsilon^2}{2}C =: C'$ uniformly in n and x (on the compact modification region all terms are bounded since $v_n \le \bar{v}$ there too). X is continuous across switches and φ' vanishes at reflection points, so $\varphi(X_{t\wedge\zeta_k}) - C't$ is a supermartingale for the switched process up to the exit time ζ_k of $[x_*, k]$; $\mathbb{E}\varphi(X_{t\wedge\zeta_k}) \le \varphi(x_0) + C't$ while $\varphi(k) \to \infty$ gives $\mathbb{P}(\zeta_k \le t) \le (\varphi(x_0) + C't)/\varphi(k) \to 0$: no explosion. □

Remark 13 (exactly where separability is and is not needed; summary of Secs. 2–5). *Not needed*: separability is not needed for the per-channel statements to survive bounded mixing (Theorem 7), nor for the conservative side at all when $h_\perp$ is bounded below and the end is at most quadratic (Theorem 12: no hypothesis on the coupling whatsoever). *Needed*: some control of channel transfer (bounded rate, or transverse boundedness below) is needed for the explosive side; Proposition 10 shows the per-channel family $\{T_n\}$—even uniformly bounded—does not determine the coupled classification once transfer is unbounded, and Proposition 11 shows the conservative side also fails, but *only* by extracting unbounded transverse energy, i.e. only for ghosts (Theorem 12). The deciding object in the coupled setting is the time-of-flight functional T_{eff} *along the actual coupled escape mechanism*—one integral per escape *path family*, no longer one integral per channel.

6 The second boundary: the channel tail, and the exact open point

The riding constructions show that in $d > 1$ the state space $\{(x, n)\}$ has, besides the countable set of channel ends at $x = \infty$, a second end: $n \to \infty$ (at co-moving x). It is itself a Feller boundary with a one-sum criterion.

Proposition 14 (tail end of the channel chain; proved (classical))**.** *Model the transverse transfer alone as a minimal birth chain on $\mathbb{N}$ with upward rates $q_n > 0$. The tail $n = \infty$ is reached in finite time a.s. iff $\sum_n 1/q_n < \infty$, and the mean climbing time is exactly $\sum_n 1/q_n$ (Feller's classical explosion criterion for birth processes). Thus the tail end carries its own "time of flight" $T_{\text{tail}} = \sum_n 1/q_n$, in exact formal analogy with $T = \int \mathrm{d}x/v$ at a spatial end*—the dictionary acquires a second leg of the same shape*, but its datum q_n comes from the coupling matrix elements, not from any integral of the potential.*

Conjecture 15 (coupled dictionary; clean statement of what remains)**.** *For the switched escape process with adapted rates $\lambda(x; n \to m)$ derived from a (possibly unbounded) coupling, the process explodes iff the combined time-of-flight functional is finite along some accessible escape route:*

$$\mathbb{P}(\zeta < \infty) > 0 \iff \inf_{\text{admissible routes } (x(t),n(t))} \Big[T_{\text{eff}}\big(x(\cdot), n(\cdot)\big) \Big] < \infty,$$

where routes carry both spatial ends (classified by T_{eff}, Lemma 9) and the tail end (classified by T_{tail}-type sums, Proposition 14). All results above are the proved instances: separable (routes are single channels), bounded rate (routes cannot ride), riding (routes computed explicitly). The general statement is a conjecture because accessibility of a route for the given *rates is a large-deviations question we have not settled.*

Remark 16 (what this says about the paper's open problem; honest). The paper's remaining gap—deficiency of the full quartic PU, i.e. whether the finite-channel indices $(2M, 2M)$ $(\lambda > 0)$ resp. $(0, 0)$ $(\lambda < 0)$ survive $M \to \infty$ (`thm:stability`, `cor:dichotomy`, `prop:directintegral`)—is the exact quantum image of the dichotomy proved here:

- $\lambda > 0$ PU: classical escape locks onto a definite kinetic direction (paper: form ratio $\to 0.9997$), i.e. bounded channel occupation along the escape; that is the regime of Theorem 7(b)/(U), where explosivity (deficiency) is stable. The stochastic analysis therefore *supports* deficiency (∞, ∞) for $\lambda > 0$, but does not prove it: the interchannel couplings are polynomial in k, not bounded, so Theorem 7 does not apply literally—the needed strengthening (subordinate-but-unbounded transfer with bounded occupation along escape) is exactly the open tail control.

- $\lambda < 0$ PU: the classical escape is channel-tail riding (paper: climbs through arbitrarily high mode-2 channels), the regime of Proposition 11/Theorem 12; there the classical and quantum legs *can* and apparently *do* split (classical blowup, quantum essential self-adjointness at every finite M, drift evidence for the full operator): zero-point transverse energy renormalizes the effective rates q_n, closing the tail. Our framework locates the split precisely at the tail end and explains its mechanism (the tail leg's datum is coupling-derived and quantization-sensitive, unlike the potential-derived T_n); it does not decide the full-operator question, which remains open, as in the paper.

Note the direction of robustness proved here (conservative side stable under arbitrary coupling for non-ghost transverse dynamics, Theorem 12; explosive side fragile, Proposition 10) matches the asymmetry of the paper's numerical verdicts: the completion candidates are both on the "classically explosive but quantum-safe" side, never the reverse.

Remark 17 (continuous transverse spectrum / sphere at infinity; conjecture). If $h_\perp$ has continuous spectrum (the direct-integral situation of the paper's `prop:directintegral`), the channel ends form a measure space of fibers. Per-fiber classification holds a.e. (Lemma 1 and the 1D Lemma fiberwise, proved), and measurable fields of lateral conditions mirror measurable fields of deficiency solutions; but individual fiber ends have zero capacity for the coupled process, so lateral data must be specified as kernels up to null sets, and the transverse operator in the fiber frame is unbounded (the paper's proposition)—the coupled classification is again governed by tail control, and we do not claim it. This, not the geometry of S^{d-1}, is the honest form of the "sphere at infinity": a direct integral of clocks, with the same open tail.

Verdict

PER-CHANNEL-ONLY (with sharp boundaries in both directions):

- **Proved, full generalization where the paper is separable/central/boundedly mixed.** The escape-diffusion dictionary extends verbatim channel by channel: the escaping end becomes the countable set of channel ends, $T(E)$ becomes $T_n = T(E - E_n)$, each end is regular iff $T_n < \infty$ (all in the same class), per-channel mean explosion time T_n^2 resp. $T_n + O(\varepsilon^2)$, per-channel speed measure = the WKB density of the paper's deficiency vector $u_n \otimes \varphi_n$, and deficiency (∞, ∞) corresponds to the infinite family of Feller–Wentzell lateral conditions with channel-mixing jump-in kernels—the stochastic S-matrix at infinity (Theorem 2; Corollary 6 with uniformity $T_\ell \propto \ell^{(2-\alpha)/(2+\alpha)}$; Theorem 7 = the stochastic Kato theorem, under the uniform time-of-flight condition (U), automatic for power-law ends, Lemma 8).

- **Proved obstruction.** "One integral per channel" is the *most* that survives: the family $\{T_n\}$ does not determine the coupled classification under unbounded channel transfer. Counterexample I (shelf potential, $\sup_n T_n \le \frac{3}{2}$, riding gives $T_{\text{eff}} = \infty$): all channels explosive, coupled

process conservative. Counterexample II (ghost transverse spectrum): all channels conservative, coupled process explosive—and this direction is provably *ghost-specific* (Theorem 12: impossible when $h_\perp$ is bounded below and the end is at most quadratic). The deciding object in genuine multi-D is the time-of-flight functional along the coupled escape route, plus a second, tail boundary at $n = \infty$ with its own one-sum criterion (Proposition 14).

- **Open / conjectural, matching the paper's open problem.** The route-infimum dictionary (Conjecture 15); the countable-boundary completeness of the lateral-condition parametrization; and the quantum tail (full PU at $M \to \infty$), where the classical and quantum legs provably can split and the stochastic leg must be run with quantization-corrected transfer rates.

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